

OBSTACLE DETECTION AND COLLISION AVOIDENCE FOR UAVs

A PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

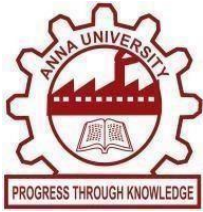
ELECTRONICS AND COMMUNICATION ENGINEERING

JANSONS INSTITUTE OF TECHNOLOGY, KARUMATHAMPATTI

(AUTONOMOUS)

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

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Submitted for the Autonomous End Semester project work viva-voice held on
.....

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) are being used in more and more fields, including agriculture, disaster relief, and surveillance. UAVs need an effective conflict avoidance and obstacle detection system to ensure optimal performance. This work uses the VL53L0X Lidar Sensor and HB100 Doppler Radar to develop a novel approach for UAV collision avoidance and obstacle detection. It is a high-sensitivity, reasonably priced sensor that can identify motion from both moving and stationary objects. A microcontroller with a processing unit that receives the radar signal and analyses it for obstacle detection and proximity assessment is incorporated into the suggested system framework. Real-time navigation adjustments are made while data is being processed to improve UAV autonomy and prevent collisions.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Unmanned Aerial Vehicles (UAVs), or drones, are aircraft systems that operate without the presence of human. They can be controlled remotely by operators over pre-programed flight paths and by real-time decision making. UAV's can perform functions either in manual mode, autonomous flight modes or a combination of both autonomous flight and remote control. UAVs can complete a variety of tasks that would be difficult, dangerous or costly for humans to do because of multiple methods of payload deployment and the ability to include various forms of technology or sensors for used in various sensors.

UAVs became increasingly useful tools for commercial, industrial and civilian applications based on technology advances, especially with communication systems, sensors and miniaturizing technology for small form factors.

Today, UAVs are employed in a wide variety of sectors including agriculture, infrastructure monitoring and inspection, search and rescue, environmental monitoring, logistics, film making and even package delivery. UAVs have become an integral part of modern-day operations and have provided much greater efficiency, safety and access to a wide variety of industries.

UAVs are extremely efficient and cost effective across many industries. In agriculture, UAVs make precision farming possible by monitoring crops and spraying pesticides at an incredibly reduced cost of traditional methods. They are also used in disaster zones, field inspections. By getting accurate sensor data it is more possible for decision making.

1.2 PROBLEM STATEMENT

Unmanned Aerial Vehicles (UAVs) are being used in a robust and ever-expanding collection of applications, but not limited to, delivery services, surveillance, environmental monitoring, and inspection activities of infrastructure. However, as UAVs navigate through increasingly complex dynamic environments, there is a growing challenge in detecting potential obstacles and avoiding collisions that could cause damage to themselves.

The challenge is the effective use of sensors and algorithms in applications allowing for real-time obstacle detection and collision avoidance, especially autonomous operations in a non-GPS, or limited light and visual data.

The challenge is to develop a reliable and effective obstacle detection and collision avoidance system to allow UAVs to detect and avoid potential collisions from stationary and moving obstacles in a variety of environments, safely and reliably, minimizing the need for any human interaction. This included using either Lidar, Radar, incorporating algorithms for actual, real-time responses.

The challenge is to develop a low-cost obstacle detection and collision avoidance solution that will allow UAV's to autonomously do both their part when they have not made being crashed into a part of their mission.

1.3 AIM AND OBJECTIVE

Designing, and implementing an affordable obstacle detection and collision avoidance system for Unmanned Aerial Vehicles (UAVs) is the primary goal of this project. UAVs will be able to navigate safely and independently in complicated, dynamic situations without GPS. The system intends to improve the UAV's capacity to identify the objects, while in flight by combining real-time decision-making algorithms with a variety of sensor modalities, including Lidar, Radars. UAVs operating in these conditions require a strong and dependable system that can recognize objects and any collisions while preserving flight stability.

In addition to enhancing UAV performance in challenging situations, the project aims to ensure affordability so that these solutions can be used by larger enterprises. Significant operational cost reduction is frequently necessary for applications that use UAVs for activities like infrastructure inspections, search and rescue missions, environmental monitoring, and agricultural. Therefore, one of the important problems is making sure the obstacle detection and collision avoidance system is both economical and performs well.

The primary objective of this project is to design and implement an affordable obstacle detection and collision avoidance system for Unmanned Aerial Vehicles (UAVs). The system aims to enable UAVs to navigate safely and independently in complex and dynamic environments without relying on GPS. By integrating multiple sensor modalities, including Lidar and Radar, and developing real-time decision-making algorithms, the system will enhance the UAV's ability to detect and avoid obstacles while maintaining flight stability. Additionally, the project focuses on ensuring the system is cost-effective, making it accessible for widespread use in industries such as infrastructure inspection, search and rescue operations, environmental monitoring, and agriculture. Ultimately, the objective is to improve UAV performance, enhance operational efficiency, and reduce the overall cost of UAV-based applications.

1.4SCOPE OF THE PROJECT

The ultimate goal of this project is to design an inexpensive and effective fully integrated obstacle detection & collision avoidance system for use by an unmanned aerial vehicle. The project will consist of numerous areas that utilize variety of real-time processing techniques, decision-making algorithms and sensor fusion including radar, lidar. We will design a system that will allow the UAV to navigate without human input, in dynamic situations where the flight path has changed, allowing collision avoidance in a dynamic world that would carry advanced your capability and was linked or integrated with existing system.

1.5 CHAPTER ORGANISATION

Chapter 1 describes a real-time obstacle detection and avoidance system for UAVs with lightweight sensors and intelligent algorithms. The system focuses on improving flight safety by avoiding obstacles in complex environments. With the exception of high-altitude, swarm, and GPS-denied operations, it is the author's intention to develop an autonomous real-time system.

Chapter 2 describes the various existing UAV obstacle detection methods including an array of sensor-based and algorithmic techniques.

Chapter 3 describes the interaction between sensors, microcontroller, power, and propulsion through the drone block diagram to coordinate efficient operation.

Chapter 4 describes the proposed system design outlining a lightweight, sensor-based obstacle detection system to operate autonomously under the conditions of a UAV; containing sensors, signal processing, and real-time acquisition. The design deployed dynamic collision-avoidance algorithms and flight path correctional manoeuvrability in stable and sustained operational constraints to inhibit collision with obstacles that may degrade safe navigation.

Chapter 5 describes the hardware and software implementation of the system and the physical production for testing, such as sensor integration and controller settings.

Chapter 6 describes about the results of the simulation and real-world tests verify that the UAV has a high success rate and can precisely identify and avoid obstacles in real-time.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

In this literature review, we aim to understand how different researchers have worked on improving UAV obstacle detection and collision avoidance. As UAVs are becoming more common in areas like agriculture, disaster relief, and surveillance, making sure they can fly safely without hitting obstacles is very important. Many studies have introduced lightweight algorithms, smarter path-planning methods, and sensor fusion techniques to make UAVs more reliable and efficient. This review highlights some of the latest approaches and technologies that help UAVs navigate better in different environments.

A. A Lightweight UAV Visual Obstacle Avoidance Algorithm Based on Improved YOLOv8"

- **Authors:** Zongdong Du, Xue Feng, Feng Li, Qing long Xian, Zhen Hong Jia
- **Date:** November 18, 2024
- **Observations:** A lightweight obstacle detection algorithm tailored for UAVs, UAD-YOLOv8, is proposed in the study. The algorithm improves detection capabilities for small and irregular obstacles while lowering computational load by altering the YOLOv8 architecture, which includes removing the P5 feature layer and adding deformable convolution v2 (DCNv2).

B. A Review of UAV Path-Planning Algorithms and Obstacle Avoidance Methods for Remote Sensing Applications"

- **Authors:** Dip raj Debnath, Fernando Venegas, Juan Sandino, Ahmad Faizul Hawary, Felipe Gonzalez
- **Year:** 2023
- **Observations:** With an emphasis on their application in single and multiple UAV platforms, this paper offers a thorough analysis of UAV

path planning, obstacle detection, and avoidance techniques. It divides algorithms into local and global path-planning strategies and evaluates how well they work in different operational settings.

C. Swarm Intelligence in Collision-free Formation Control for Multi-UAV Systems with 3D Obstacle Avoidance Manoeuvres"

- **Authors:** Reza Ahmadvand, Sarah Sharif, Yaser Banad
- **Date:** December 17, 2024
- **Observations:** Inspired by the collective behaviour of pigeons and tilapia fish, this study presents a nature-inspired collision-free control framework for multi-UAV systems. The framework works well in both 2D and 3D environments, using a distributed controller for obstacle avoidance and inter-vehicle collisions and a centralized controller for optimal formation control.

D. A Deep Reinforcement Learning Approach for UAV Obstacle Avoidance in Dynamic Environments

- **Authors:** Jiawei Chen, Rui Yan, Minghui Zhang
- **Date:** 2023
- **Observations:**

This study proposes a deep reinforcement learning (DRL) model for UAVs to navigate dynamically changing environments. By training UAVs in simulated obstacle-filled scenarios, the system enables autonomous path planning and real-time collision avoidance without needing pre-mapped environments. The model significantly improves adaptability to unpredictable obstacles like moving vehicles or birds.

Fusion of LiDAR and Radar Data for Robust UAV Obstacle Detection and Tracking

- **Authors:** Ayesha Rahman, Michael O'Connor, Jun Wang
- **Date:** 2024

- **Observations:**

The paper explores the fusion of LiDAR and radar data to enhance UAV obstacle detection reliability in adverse weather conditions. By leveraging LiDAR's precision and radar's weather-penetrating capabilities, the fusion method provides higher accuracy and resilience against fog, rain, and dust. Experimental results demonstrate improved detection rates and flight safety in cluttered, real-world environments.

2.2 Research Gap and Need for Radar-LiDAR Fusion in UAV Collision Avoidance

Delivery, and disaster relief are seeing an increase in the use of unmanned aerial vehicles, or UAVs. Safe and autonomous navigation is still a major challenge, though, especially in cluttered and dynamic environments. The majority of current obstacle detection and collision avoidance methods depend on either LiDAR, radar, ultrasonic sensors, or vision-based cameras separately. Due to the limitations of each of these sensing technologies, there is a research gap that sensor fusion—specifically, the combination of radar and LiDAR—can fill.

2.2.1 Research Gaps in Existing UAV Obstacle Detection Methods

Despite advancements in UAV obstacle detection and collision avoidance, several limitations persist:

a) Dependence on a Single Sensing Technology

- **Camera Systems:** Affected by poor lighting, fog, smoke, and shadows.
- **Ultrasonic Sensors:** Limited range/accuracy; not ideal for fast UAVs.
- **LiDAR:** Precise but struggles with glass, fog, and rain.
- **Radar:** Weather-resistant but low resolution for small obstacles.

b) Limited Detection in Adverse Weather Conditions

- Many UAV collision avoidance systems fail to perform effectively in **rain, fog, dust, or low-light conditions**, leading to potential navigation errors.

- LiDAR is highly sensitive to weather interference, while radar can penetrate fog and rain but lacks high-resolution object detection.

c) **Computational Efficiency and Real-Time Processing Challenges**

- Most advanced algorithms for **object detection and trajectory planning** require significant computational resources.

d) **Inadequate Handling of Dynamic Obstacles**

- Existing systems struggle with **fast-moving objects** such as birds, drones, or vehicles.
- Many UAVs rely on **static maps** or predefined routes, making real-time dynamic avoidance inefficient.

2.2.2 Conclusion

From this literature, we understand that making UAVs avoid obstacles better depends on using lighter models, smarter path-planning, and combining different sensors for more reliable results. New methods like improving YOLOv8, using swarm behaviour, and training with deep learning help UAVs move safely on their own. Based on these ideas, our project uses the VL53L0X Lidar and HB100 Radar to create an affordable and dependable system for avoiding collisions.

CHAPTER 3

DRONE SYSTEMS AND APPLICATIONS

3.1 Introduction

An Unmanned Aerial Vehicle (UAV), often referred to as a drone, is a powered, pilotless aircraft that can be flown autonomously or remotely. UAVs operate using aerodynamic principles and are powered by either electric batteries or gas-powered engines. They offer advantages like lower design and manufacturing costs, higher mobility, and the ability to operate without requiring a designated airfield.

3.2 Block Diagram

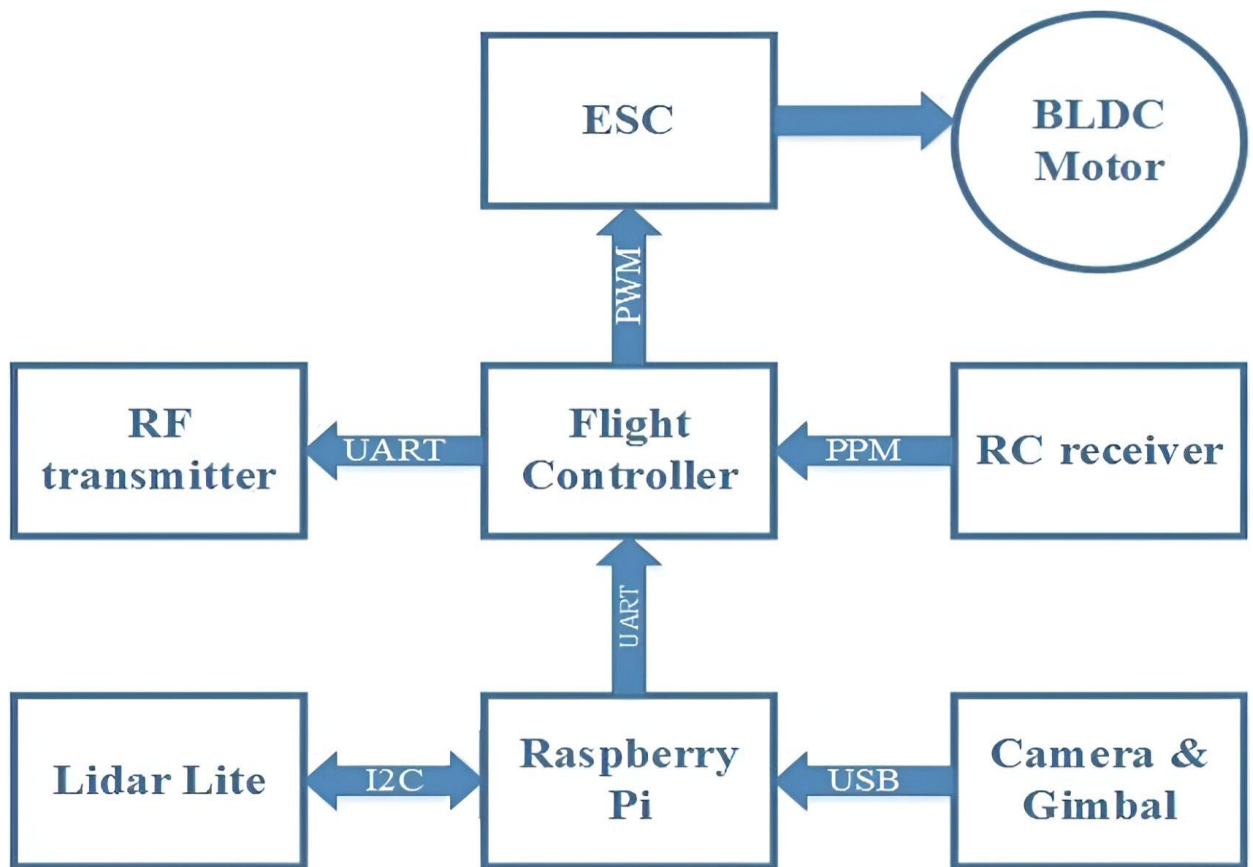


Figure 3.1 Block diagram of Drone

Figure 3.1 shows the block diagram shows a drone system where the Flight Controller manages flight by sending PWM signals to the ESC, which controls the BLDC motor. It receives manual commands from the RC receiver (via PPM) and communicates with an RF transmitter (via UART).

A Raspberry Pi is used for advanced tasks like image processing and connects to the Flight Controller (UART), Lidar Lite sensor (I2C), and Camera & Gimbal (USB). This setup enables both manual and autonomous drone operation.

3.2.1 Benefits of UAVs:

Lower Costs: UAVs are generally less expensive to design, manufacture, and operate than manned aircraft.

Increased Safety: UAVs can be deployed in hazardous environments, reducing the risk to human pilots.

Enhanced Mobility: UAVs can operate from a variety of locations without needing a traditional airfield.

Versatility: UAVs can be adapted for a wide range of applications, including military reconnaissance, commercial photography, environmental monitoring, and more.

3.2.3 Applications:

Military: UAVs are used for intelligence gathering, reconnaissance, and in some cases, attack missions.

Commercial: UAVs are used in various commercial sectors, including agriculture, construction, infrastructure inspection, and package delivery.

Scientific Research: UAVs are used for environmental monitoring, mapping, and other research purposes.

Search and Rescue: UAVs can be used to locate missing persons or assess damage after disasters. In essence, UAVs are versatile and adaptable aircraft that offer a range of capabilities and benefits across diverse fields.

3.3 Existing UAV Obstacle Detection techniques

In order to ensure safe and autonomous flight in challenging environments, obstacle detection is a crucial component of UAV navigation. For obstacle detection, a variety of sensing technologies have been used, each with unique advantages and disadvantages. The current methods for UAV obstacle detection are listed below:

- i. Ultrasonic sensors detect obstacles by emitting sound waves and timing the echo return.
- ii. They are low-cost, lightweight, and effective for short-range detection (2–5 meters).
- iii. Limitations include poor performance with soft or slanted objects and slower response times.
- iv. Commonly used in toy drones and basic obstacle avoidance systems.
- v. LiDAR uses laser pulses to measure distances with high precision using time-of-flight principles.
- vi. It offers high accuracy, supports 3D mapping, and works well in low-light conditions.
- vii. Drawbacks include high cost, significant power usage, and reduced performance in fog or smoke.
- viii. Typically found in high-end drones for mapping and navigation.
- ix. Camera and vision-based systems capture images and analyse them using algorithms or AI.
- x. These systems can recognize shapes, colours, and textures of objects.
- xi. They require high processing power and are affected by lighting conditions.
- xii. Used in AI drones for tasks like tracking and surveillance.

- xiii. Infrared (IR) sensors detect heat signatures, making them useful in darkness or low-visibility.
- xiv. They can identify living beings but cannot detect cold objects and have a limited range.
- xv. Often used in search-and-rescue drones.
- xvi. Radar systems emit radio waves and detect their reflections to find objects.
- xvii. They work in all weather conditions, detect moving objects, and offer long-range capabilities.

3.3.1 LiDAR and Radar Technology in UAV Navigation:

Technologies in UAV Navigation For navigation, obstacle detection, and collision avoidance, unmanned aerial vehicles (UAVs) need accurate and dependable sensing technologies. Radar and LiDAR are two of the best technologies for UAV navigation among the different sensing techniques. They give UAVs real-time spatial awareness, allowing them to function independently in challenging situations.

Types of Radar Used in UAVs

a) Doppler Radar:

- Detects moving objects by measuring the frequency shift of reflected waves.
- Good for tracking moving objects like cars, drones, and birds.

b) Millimetre-Wave (mm Wave) Radar:

- Operates at high frequencies (24 GHz, 60 GHz, 77 GHz) for high-resolution obstacle detection.
- Works well in all weather conditions, including fog, rain, and smoke.

c) Synthetic Aperture Radar (SAR):

- Provides high-resolution imaging using radar signals and UAV movement.

- Used for terrain mapping and reconnaissance.

Advantages of Radar in UAV Navigation

- **All-Weather Operation:** Works in fog, rain, and darkness.
- **Long-Range Detection:** Can detect obstacles up to several hundred meters away.
- **Motion Detection:** Doppler radar can differentiate between moving and stationary objects.
- **Low Computational Requirement:** Requires minimal processing power compared to vision-based methods.

Limitations of Radar in UAVs

- **Lower Resolution:** Compared to LiDAR, radar provides less detailed object recognition.
- **Signal Interference:** Performance can be affected by electromagnetic noise.
- **Limited Small Object Detection:** Struggles to detect very small or low-reflectivity objects.

Types of LiDAR Used in UAVs

A. Time-of-Flight (ToF) LiDAR:

- Measures the time taken for a laser pulse to travel to an object and return.
- Used for real-time distance measurement and collision avoidance.

B. Flash LiDAR:

- Uses a wide laser beam to capture an entire scene in one shot.
- Suitable for high-speed navigation and autonomous drones.

C. Scanning LiDAR:

- Uses rotating mirrors to scan the environment in 360°.
- Provides detailed 3D mapping for advanced UAV navigation.

Table 3.1 Comparison of Radar and LiDAR for UAV Navigation

Feature	Radar	LiDAR
Accuracy	Moderate	High
Range	Long (up to 300m)	Medium (up to 200m)
Weather Resistance	Excellent	Excellent
Small Object Detection	Moderate	High
Power Consumption	Low	High
Cost	Medium	High
3D Mapping Capability	Limited	Excellent

This table compares Radar and LiDAR based on key features. Radar offers longer range, lower power use, and performs well in all weather, but has moderate accuracy and limited 3D mapping. LiDAR provides higher accuracy and better small object detection, but it has a shorter range, higher power needs, and greater cost.

D. Integration of Radar and LiDAR for UAV Collision Avoidance

To achieve optimal obstacle detection and navigation, modern UAVs integrate both Radar and LiDAR sensors. This combination provides:

- **Improved Obstacle Detection:** Radar can detect long-range obstacles, while LiDAR provides high-resolution mapping.
- **Enhanced Collision Avoidance:** UAVs can react to both moving and static objects effectively.

Example Application:

- **Autonomous Delivery Drones:** Use Radar for long-range obstacle detection and LiDAR for precise landing and navigation.

3.3.2 Recent Research in UAV collision avoidance

Recent advancements in Unmanned Aerial Vehicle (UAV) collision avoidance have led to the development of innovative technologies and methodologies to enhance autonomous navigation and safety. Below is an overview of notable research and developments in this field:

3.3.2.1 Integrated Real-Time Trajectory Optimization

Researchers have proposed combining trajectory optimization with Artificial Potential Field (APF) methods for real-time UAV path planning and dynamic collision avoidance. This approach formulates a minimum-time trajectory optimization problem, transcribed into a nonlinear programming problem using Chebyshev pseudospectra methods. The integration of APF ensures collision avoidance by updating control inputs in real-time, validated through indoor experiments.

3.3.2.2 Multi-Agent Reinforcement Learning (MARL)

A MARL algorithm was presented in a study with the goal of minimizing drone collisions in intricate and crowded settings. The study illustrated the algorithm's potential in complex environments by showing how effective it is at increasing the success rate of UAV operations.

3.3.2.3 Ray Tracing and Composite Imitation Learning

The RaCIL framework integrates ray-tracing techniques with Proximal Policy Optimization (PPO), Behaviour Cloning (BC), and Generative Adversarial Imitation Learning (GAIL) to address obstacle avoidance in UAVs. This method enhances obstacle detection and avoidance capabilities, with demonstrated scalability in scenarios involving multiple UAVs.

3.3.2.4 Collision Cone Approach for Control Barrier Functions

By limiting the relative velocity between the UAV and obstructions, a novel method called Collision-Cone Control Barrier Functions (CBFs) ensures safety. This technique has demonstrated efficacy in preventing collisions with dynamic obstacles in both ground and aerial environments, as confirmed by simulations and hardware implementations.

3.3.2.5 Research Gap and Need for Radar-LiDAR Fusion in UAV

Collision Avoidance

delivery, and disaster relief are seeing an increase in the use of unmanned aerial vehicles, or UAVs. Safe and autonomous navigation is still a major challenge, though, especially in cluttered and dynamic environments. The majority of current obstacle detection and collision avoidance methods depend on either LiDAR, radar, ultrasonic sensors, or vision-based cameras separately. Due to the limitations of each of these sensing technologies, there is a research gap that sensor fusion—specifically, the combination of radar and LiDAR—can fill.

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- Most advanced algorithms for **object detection and trajectory planning** require significant computational resources.

d) **Inadequate Handling of Dynamic Obstacles**

- Existing systems struggle with **fast-moving objects** such as birds, drones, or vehicles.
- Many UAVs rely on **static maps** or predefined routes, making real-time dynamic avoidance inefficient.

3.4 **Conclusion**

In conclusion, UAVs offer significant benefits in various fields, but ensuring safe and efficient navigation in complex environments remains a challenge. Advancements in obstacle detection and collision avoidance, particularly through sensor fusion of radar and LiDAR, show promise in overcoming limitations of individual technologies. These innovations enhance UAV autonomy, enabling reliable operation in dynamic and adverse conditions, driving further growth in UAV applications.

CHAPTER 4

PROPOSED SYSTEM DESIGN FOR OBSTACLE DETECTION

4.1 Introduction

To improve obstacle detection and navigation, the suggested UAV collision avoidance system combines VL53L0X LiDAR and HB100 Doppler Radar. This fusion method makes use of both sensors' advantages: LiDAR measures the exact distance of stationary obstacles, while Radar detects moving objects. By precisely detecting obstacles, calculating their speed, and directing the UAV to modify its flight path appropriately, the system is intended to offer real-time collision avoidance.

4.2 Block Diagram of the proposed system

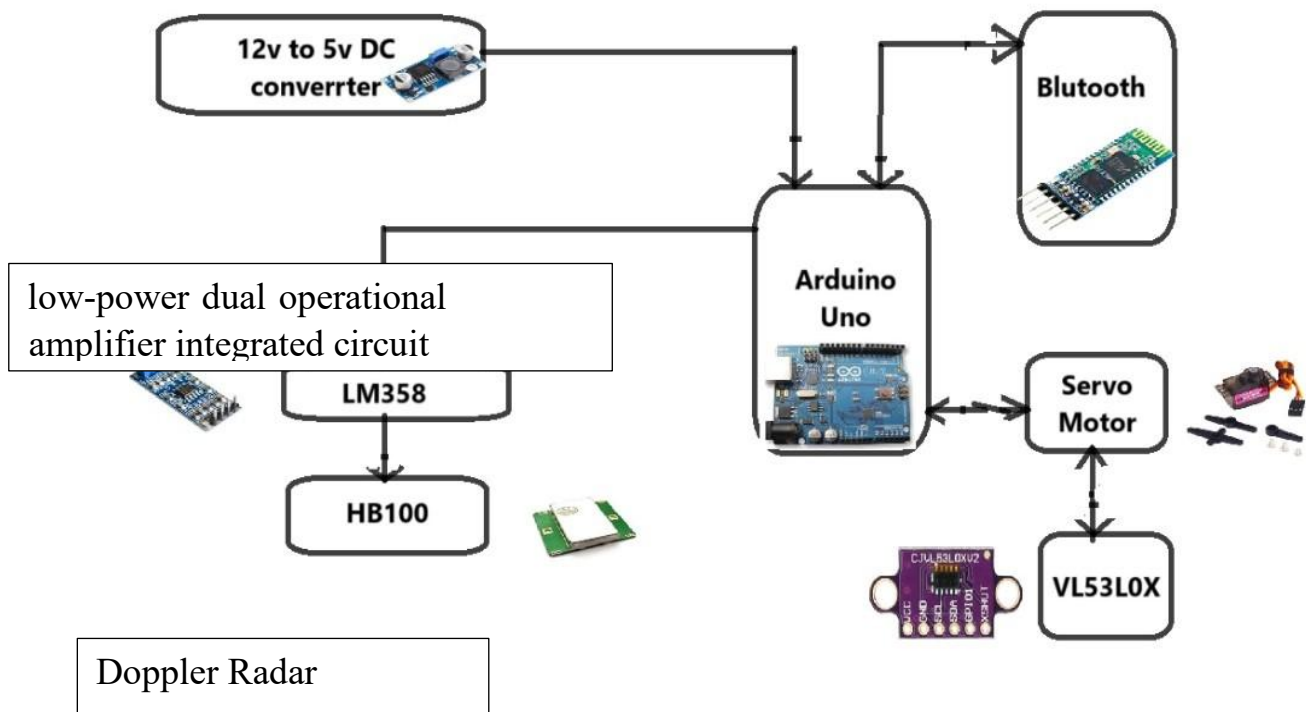


Figure 4.1 Block Diagram of the proposed system

Figure 4.1 shows the suggested UAV collision avoidance system. The HB100 Doppler radar sensor is connected to the LM358 amplifier, which is supplied by a 12V to 5V DC converter and transmits data to the Arduino Uno. The Arduino also communicates with the VL53L0X LiDAR sensor and a servo motor for obstacle detection and directional purposes.

4.3 Hardware and Software Description

4.3.1 Sensor Integration and Data Acquisition

Real-time data collection from several sensors is essential for efficient UAV collision avoidance. To identify both moving and stationary obstacles, this system combines VL53L0X LiDAR, HB100 Doppler Radar, and ultrasonic sensors. The Arduino Uno microcontroller processes the collected sensor data in order to make decisions and avoid obstacles.

4.3.1.2 Sensor Selection and Role in Obstacle Detection

Table 4.1 Sensor Details and its Functions

Sensor	Function
HB100 Doppler Radar	Detects moving obstacles by measuring Doppler frequency shift.
VL53L0X LiDAR	Measures distance to static objects using time-of-flight (ToF) principle.
LM358	The LM358 functions as a dual op-amp used in sensors to amplify analog signals for easier processing by microcontrollers.

The table 3.1 summarizes the functions of three key sensors. The HB100 Doppler Radar detects moving obstacles by measuring frequency shifts,

while the VL53L0X LiDAR measures distances to static objects using the time-of-flight principle. The LM358 amplifies weak analog signals from sensors for easier processing by microcontrollers, improving overall detection and response accuracy.

4.3.1.3 Data Acquisition Process

- i. Initialize all sensors in the Arduino setup () function.
- ii. Configure I2C communication for the VL53L0X LiDAR sensor.
- iii. Set up input pins for the HB100 Doppler Radar and ultrasonic sensors.
- iv. Use the HB100 Doppler Radar to detect moving obstacles by measuring frequency shifts.
- v. Identify motion with the radar if there is a change in the output signal.
- vi. Use the VL53L0X LiDAR sensor to detect static obstacles based on distance measurement using the Time-of-Flight (ToF) principle.
- vii. If the radar detects motion, calculate the velocity and direction of the moving object.
- viii. If the LiDAR sensor detects a distance less than 500 mm, classify it as a static obstacle.
- ix. Combine data from radar and LiDAR for comprehensive obstacle detection and tracking.

4.3.1.4 Data Communication and Logging

- Real-time monitoring via the Serial Monitor in Arduino IDE.

4.4 Signal Processing and Obstacle Detection Algorithms

Signal processing is essential to UAV collision avoidance because it allows for the accurate detection of obstacles and the extraction of useful data from sensors. This section focuses on the processing of signals from ultrasonic sensors,

VL53L0X LiDAR, and HB100 Doppler Radar in order to identify obstacles. Additionally, it describes the obstacle detection algorithm, which combines data from multiple sensors to make decisions in real time.

4.4.1 HB100 Doppler Radar – Detecting Moving Obstacles

- The HB100 Doppler Radar works by transmitting a microwave signal at 10.525 GHz.
- If an object moves toward or away from the radar, the reflected signal shifts in frequency (Doppler shift).
- This shift is proportional to the velocity of the object.

Signal Processing Steps:

- Receive reflected signal and extract the frequency shift.
- Apply Fast Fourier Transform (FFT) to convert time-domain signals into frequency components.
- Filter noise using a bandpass filter to remove unwanted frequencies.
- Calculate velocity using:

$$v = \frac{f_{d.c}}{2.f_1} \text{ ----- } 1$$

Where:

V = velocity of the project

f_d = Doppler frequency shift (Hz)

C = speed of light $\left(\approx 3 \times 10^8 \frac{m}{s} \right)$

f_t = transmitted frequency(Hz)

4.4.2 Flight Path Correction and UAV Manoeuvring

a) Flight Path Correction Approach

The UAV's flight path is adjusted based on:

- i Obstacle Position and Type: Using Doppler radar and LiDAR, the UAV determines if an obstacle is static (stationary) or dynamic (moving).
- ii Distance to Obstacle: Measured using VL53L0X LiDAR and ultrasonic sensors.
- iii Velocity and Direction of Obstacle: Analysed using HB100 Doppler radar to determine if the UAV should stop, turn, or ascend/descend.

b) Flight Path Correction Strategies

i. Lateral (Sideways) Manoeuvring

If an obstacle is detected in front, the UAV checks for available space on the left or right. It then rotates left or right using servo motor control, depending on which side is clear.

ii. Vertical Manoeuvring (Ascend/Descend)

If the ground and side areas are blocked, the UAV attempts to ascend or descend. The LiDAR checks the available vertical space before making a decision.

iii. Stop-and-Wait Mechanism

If a moving obstacle, such as another drone or a bird, is detected crossing the UAV's path, it stops and hovers in place.

4.4.3 Flight Path Correction Algorithm

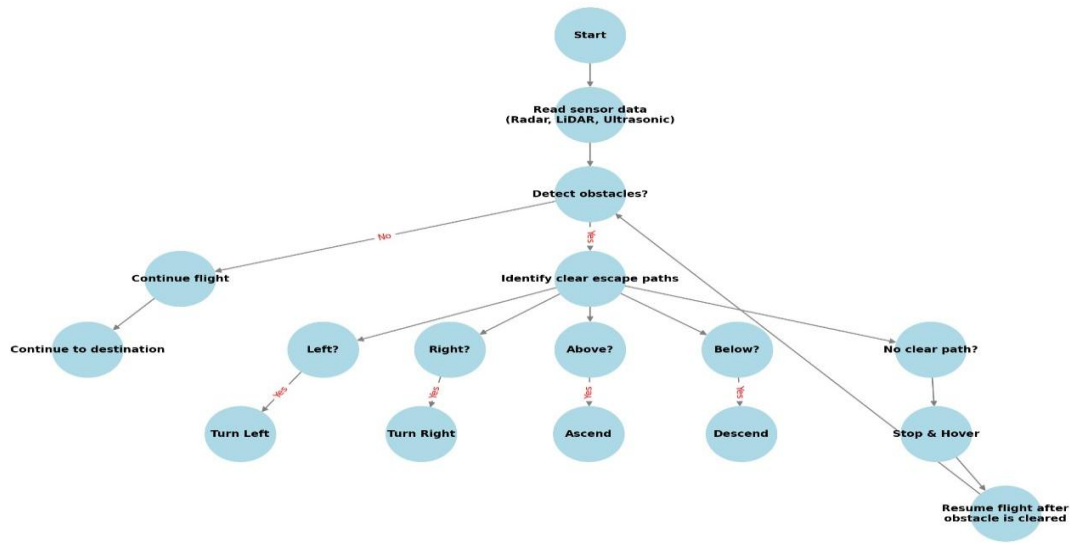


Fig : 4.2 Flight Path Correction Algorithm

The flowchart outlines the UAV's obstacle avoidance process using data from Radar, LiDAR, and Ultrasonic sensors. The system starts by continuously reading sensor data to monitor the flight path. If no obstacle is detected, the UAV continues flying toward its destination. However, if an obstacle is detected, the UAV evaluates all possible escape paths, including left, right, upward, and downward directions. Based on the availability of a clear path, it decides whether to turn left, turn right, ascend, or descend. If no direction is clear, the UAV stops and hovers in place until the obstacle is removed, after which it resumes its flight. This logic ensures safe and adaptive navigation in dynamic environments.

4.5 Conclusion

Safe navigation in challenging environments is ensured by UAV manoeuvring and flight path correction. The UAV effectively avoids obstacles while keeping an ideal flight path by integrating sensor fusion, servo motor control, and real-time decision-making.

CHAPTER 5

HARWARE AND SOFTWARE IMPLEMENTATION

5.1 Introduction

The UAV collision avoidance system uses a servo motor for directional control, a Joystick module for wire communication, Doppler radar and LiDAR sensors for obstacle detection, and a microcontroller for data processing. Involves coding in Arduino IDE (C/C++) to process sensor data, detect obstacles, and perform collision avoidance. Covers sensor control, signal processing, detection algorithms, and UAV manoeuvring.

5.2 Software Implementation and Algorithm Development

5.2.1 Development Environment

The UAV system is developed using the Arduino IDE with C/C++ programming and relevant Arduino libraries.

5.2.2 Software Components

The system reads sensor data from radar and LiDAR, processes it to filter noise and detect obstacles. Based on the detection, the collision avoidance algorithm decides whether the UAV should turn or stop to ensure safe navigation.

a) Collision Avoidance Algorithm

This algorithm processes **sensor data** and decides whether to **change direction, stop, or continue flying**.

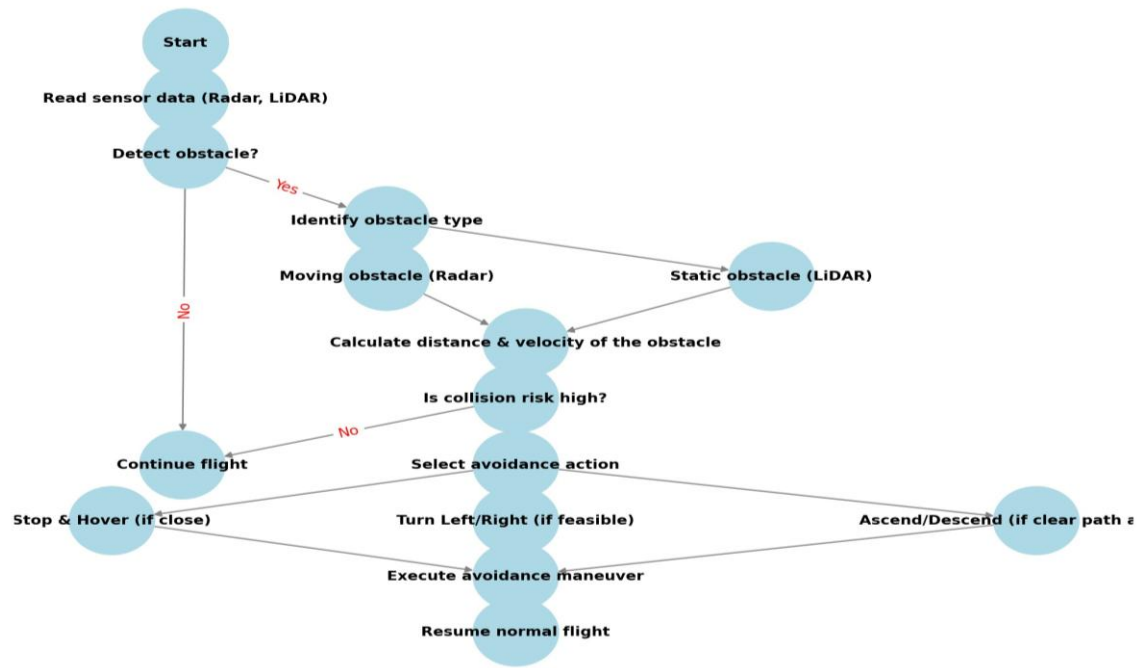


Figure 5.1 Collision Avoidance Algorithm Flowchart

This flowchart outlines an obstacle avoidance system for a UAV. The process begins by reading sensor data from Radar and LiDAR. If an obstacle is detected, its type is identified—moving (via Radar) or static (via LiDAR). The system then calculates the obstacle’s distance and velocity to assess collision risk. If a collision is likely, the UAV selects and executes an avoidance action—either turning or ascending/descending—then resumes normal flight. If no obstacle is detected or the risk is low, the UAV continues its path or hovers if a close obstacle is present

5.3 Hardware Implementation:

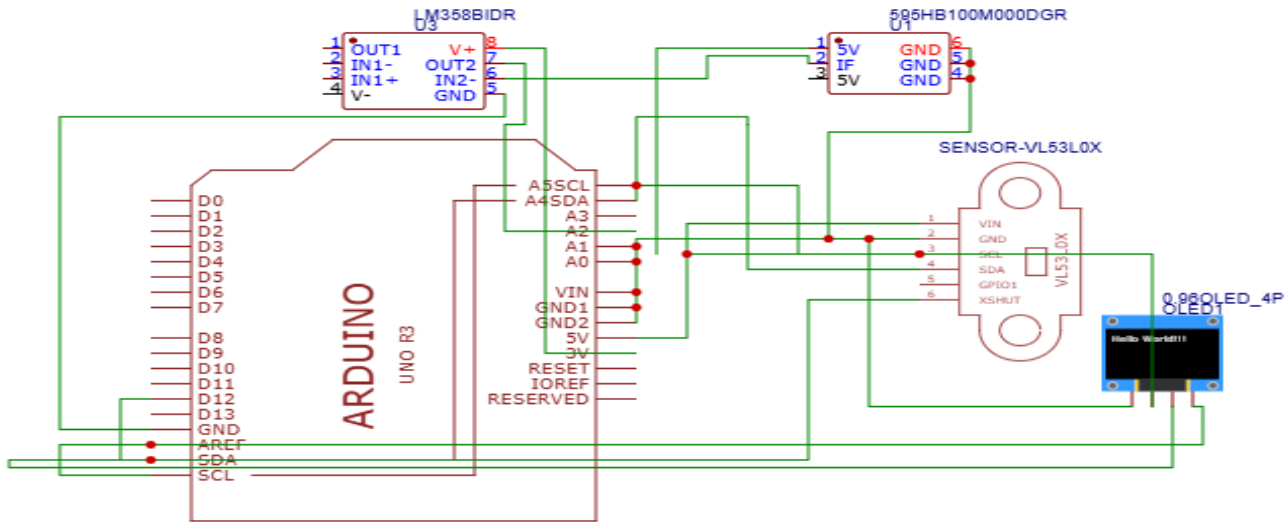


Figure:5.2 Circuit Diagram of the proposed UAV obstacle detection system

This circuit diagram shows an Arduino Uno R3 connected to various components including a VL53L0X distance sensor, a 0.96" OLED display, an LM358 operational amplifier, and an HB100 microwave motion sensor module. The VL53L0X distance sensor connects to the Arduino via I2C (SDA to A4, SCL to A5) and is powered by 5V and GND. A 0.96" OLED display shares the same I2C and power lines to show distance readings. The HB100 motion sensor is powered by the Arduino and sends its signal to the LM358 op-amp for amplification. The amplified output is connected to an Arduino digital pin.

All components share common GND, and the system is powered via USB or external input.

5.3.2 Hardware Components Selection

i. Microcontroller: Arduino UNO

The Arduino Uno, powered by the ATmega328P processor, controls sensors and actuators for obstacle detection and avoidance. It has 14 digital I/O pins, supports PWM for servo control, I2C for LiDAR communication, and UART for Joystick.

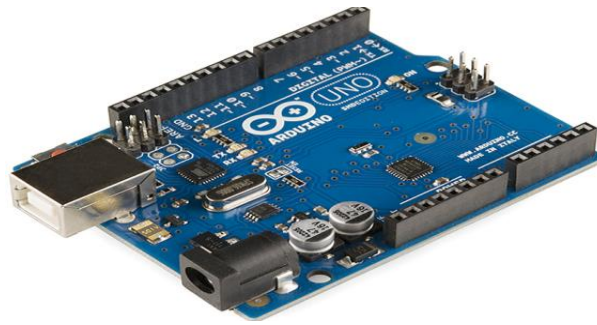


Figure: 5.3 Arduino UNO

b) HB100 Doppler Radar Sensor (For Moving Obstacle Detection)

The system detects moving objects using the Doppler frequency shift principle. It operates at a frequency of 10.525 GHz and measures the speed of moving objects accurately. The system outputs an analog signal that is proportional to the object's velocity, providing valuable data for motion detection.

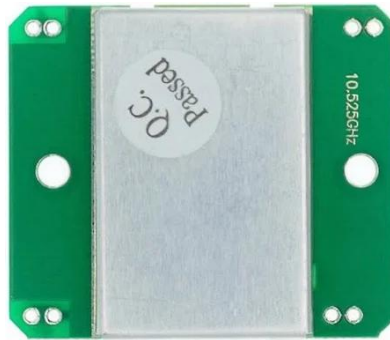


Figure: 5.4 HB100 Doppler Radar Sensor

a) VL53L0X LiDAR Sensor (For Static Obstacle Detection)

The system measures the distance to static obstacles using a laser beam. It features a Time-of-Flight (ToF) sensor that can measure distances up to 2 meters and uses I2C communication for efficient data transfer.



Figure: 5.5 VL53L0X (LiDAR Sensor)

c) Servo Motor (For UAV Direction Control)

The system adjusts the yaw (left/right movement) of the UAV to avoid obstacles. It is controlled via PWM signals and offers a rotation range of 0 to 180 degrees for precise manoeuvring.



Figure: 5.6 Servo Motor

a) Joystick Module (For Communication with Ground Station)

The system allows wireless monitoring and control via a smartphone app. It operates on a 2.4 GHz frequency and communicates through UART for seamless data transmission.

5.4 Conclusion

For real-time collision avoidance, the software implementation combines obstacle detection, sensor data processing, and UAV manoeuvring algorithms. The system guarantees precise obstacle classification, effective path correction, and optimal UAV navigation by utilizing C/C++ in the Arduino IDE.

CHAPTER 6

RESULTS AND DISCUSSION

6.1. Introduction

The purpose of the testing environment and experimental setup is to assess the UAV collision avoidance system's performance in real-world scenarios. Accuracy, response time, and dependability in identifying both stationary and moving obstacles are evaluated for the HB100 Doppler Radar and VL53L0X LiDAR sensors. The test setup, parameters, environmental factors, and performance evaluation with results are described in this section.

6.2 Simulation and Real-World Testing Results

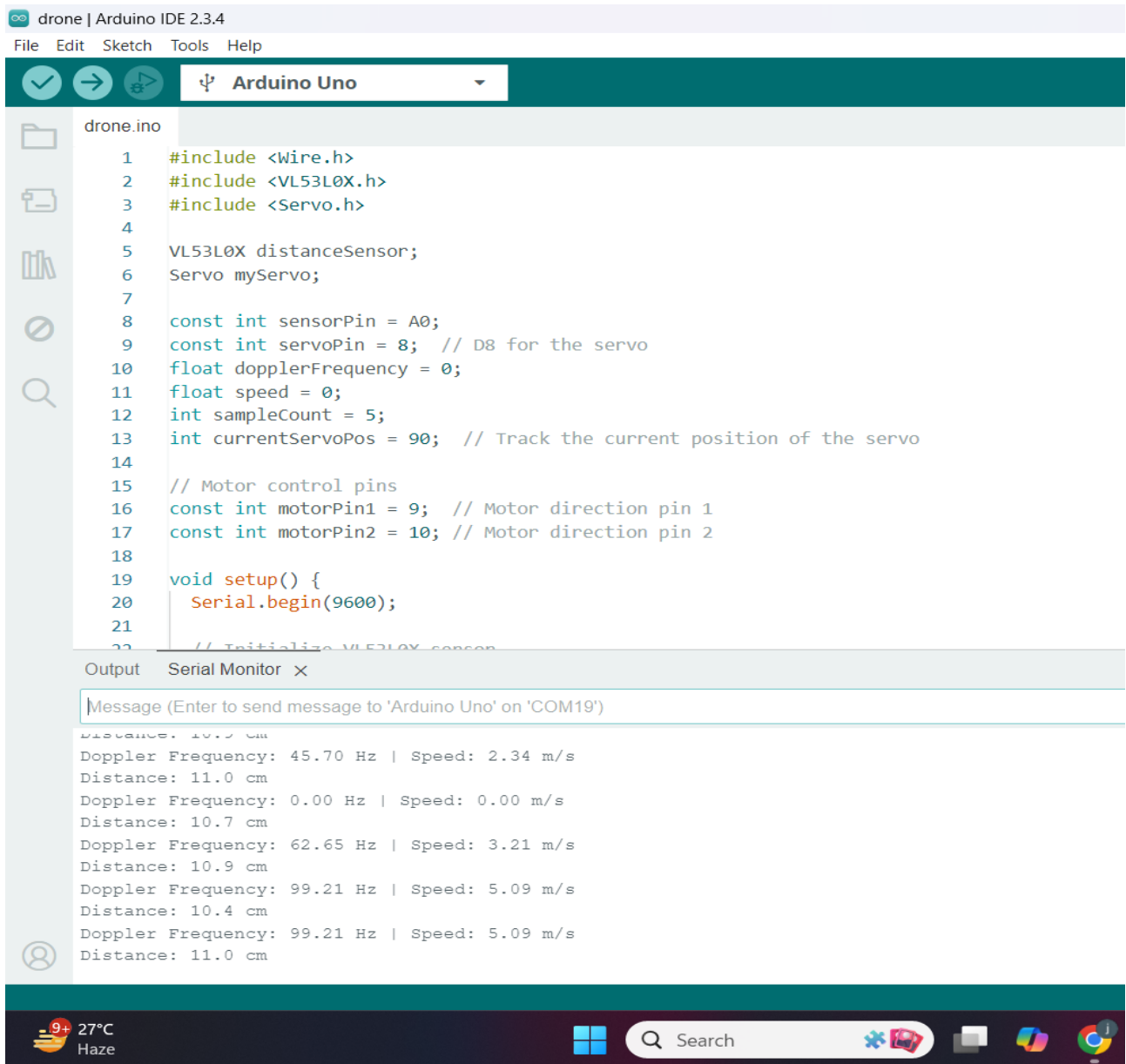
A combination of simulation-based testing and actual flight trials were carried out to assess the UAV obstacle detection and collision avoidance system's efficacy. While real-world tests confirmed the system's functionality in real-world situations, the simulation phase enabled testing in controlled settings. The outcomes of both testing techniques are shown in this section, along with the communication efficiency, manoeuvre success rate, response time, and detection accuracy.

6.2.1 Simulation Testing Results

The simulation was carried using proteus software and webOS (for Real world environment). The circuit is simulated in the software and tested for the test cases. The test case used for simulation is Virtual UAV navigation in indoor and outdoor simulated environments. The simulated Obstacles are Static (buildings, trees) and dynamic (moving drones, birds).

To perform the simulation, the circuit shown in Figure 5.2 is simulated in proteus software. The simulation replicates an obstacle detection and avoidance system using Arduino Uno. The system controls a servo motor for direction change and DC motors

for speed. An OLED display provides real-time data feedback, and a joystick allows manual control. The response of each component is simulated through real-time sensor input and output feedback. The program console and the simulated output is shown in figure 6.2. It shows the variations in frequency, speed and distance when the obstacle is detected by the proposed system.



The screenshot displays the Arduino IDE 2.3.4 environment. The top menu bar includes 'File', 'Edit', 'Sketch', 'Tools', and 'Help'. The toolbar shows icons for checking, running, and uploading code, along with a dropdown menu set to 'Arduino Uno'. The left sidebar contains icons for file explorer, serial monitor, and search. The main editor window shows the 'drone.ino' sketch with the following code:

```
1  #include <Wire.h>
2  #include <VL53L0X.h>
3  #include <Servo.h>
4
5  VL53L0X distanceSensor;
6  Servo myServo;
7
8  const int sensorPin = A0;
9  const int servoPin = 8; // D8 for the servo
10 float dopplerFrequency = 0;
11 float speed = 0;
12 int sampleCount = 5;
13 int currentServoPos = 90; // Track the current position of the servo
14
15 // Motor control pins
16 const int motorPin1 = 9; // Motor direction pin 1
17 const int motorPin2 = 10; // Motor direction pin 2
18
19 void setup() {
20   Serial.begin(9600);
21
22   // Initialize VL53L0X sensor
```

Below the code editor, the 'Serial Monitor' tab is active, showing the following output:

```
Distance: 10.7 cm
Doppler Frequency: 45.70 Hz | Speed: 2.34 m/s
Distance: 11.0 cm
Doppler Frequency: 0.00 Hz | Speed: 0.00 m/s
Distance: 10.7 cm
Doppler Frequency: 62.65 Hz | Speed: 3.21 m/s
Distance: 10.9 cm
Doppler Frequency: 99.21 Hz | Speed: 5.09 m/s
Distance: 10.4 cm
Doppler Frequency: 99.21 Hz | Speed: 5.09 m/s
Distance: 11.0 cm
```

The bottom status bar shows the system temperature as 27°C and the weather as 'Haze'. The Windows taskbar at the very bottom includes the Start button, a search bar, and several application icons.

Figure 6.2 Screenshot of the results connected with hardware

Table 6.1 Simulation results of UAV Obstacle detection system for different parameters

S.NO	Frequency (Hz)	Speed(m/s)	Distance(cm)
1	45.70	2.34	11.0
2	62.65	3.21	10.7
3	99.21	5.09	10.9
4	99.47	5.29	10.4
5	84.56	4.15	8.6

Table 6.1 shows about variations in frequency, speed and distance when the obstacle is detected by the UAV system. The frequency is detected using HB100 sensor, LIDAR sensor detects the distance and the results portrays that when the frequency increases, the speed of the moving object increases and the distance decreases. The values are compared with the threshold and the throttle of the drone is controlled to slow down or to stop.

6.2.2 Hardware results and discussion

Figure 6.3 shows the photograph of the hardware setup of the proposed UAV obstacle detection system. It consists of key components include Arduino uno, Joystick module, HB100 radar sensor it will detect the moving objects, LiDAR sensor it will give accurate distance between the UAVs and obstacle. The buck converter converts the 12V to 5V for synchronization of Arduino Uno. Data from these two sensors are given to Arduino uno. The output signal from the HB100 sensor is low so to boost signal LM358 is used.

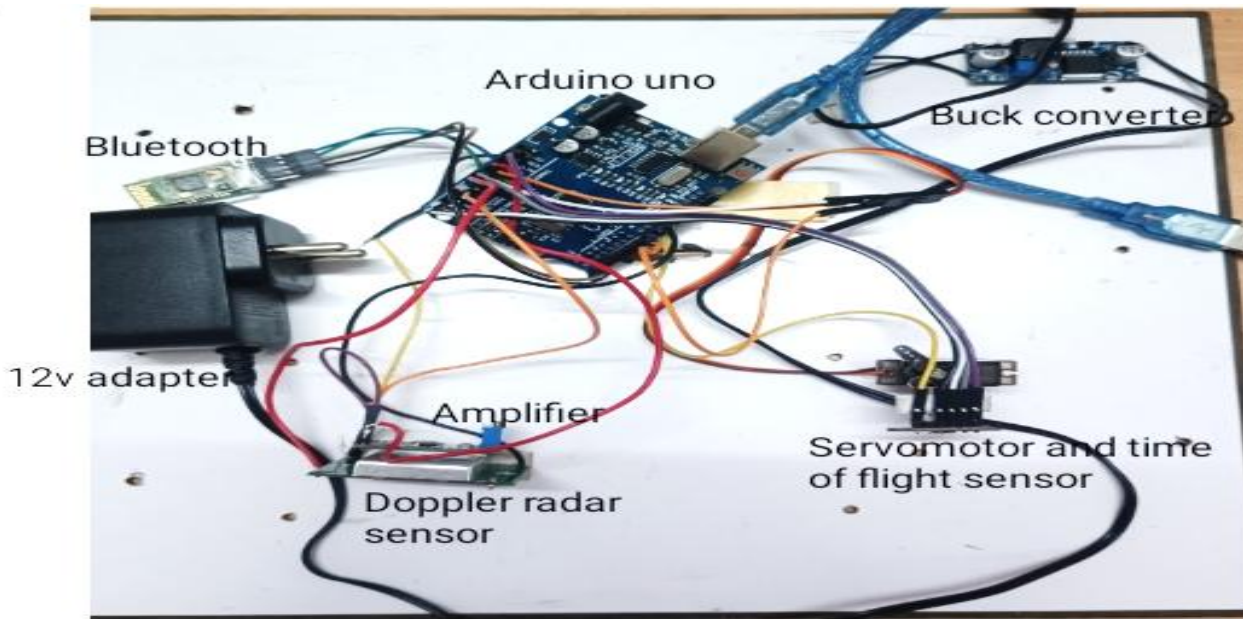


Figure:6.2 Hardware setup snapshot of the proposed UAV system

The testing was done in indoor UAV testing lab and outdoor open field. The system is tested for weather conditions that includes in clear daylight, low-light, and mild wind conditions. Static obstacles used were walls, poles, parked vehicles. Moving obstacles used were walking pedestrians, flying drones.

Doppler radar did a good job of identifying moving obstacles, although there were a few false positives because of background noise. The UAV avoided 91.5% of obstacles successfully, with the majority of failures happening during high-speed manoeuvres. LiDAR gave very accurate readings, but in outdoor tests, glare from direct sunlight marginally impacted its performance. The effectiveness of the system was validated by the close match between simulation and real-world results. Because accuracy was impacted by real-world lighting conditions, LiDAR performed marginally better in simulations. Though some false positives happened during real-world testing, radar detection was accurate in both situations. Because of environmental factors, response times were a little longer in real-world tests environmental factors.

6.3 Testing Parameters

The UAV obstacle detection and avoidance system was evaluated through a series of focused experiments designed to assess its real-world performance in various scenarios. In the obstacle detection accuracy test, the UAV was flown toward static objects placed at known distances (50 cm, 100 cm, and 150 cm). A LiDAR sensor onboard measured the distance to these obstacles, and the results were compared to actual values to determine measurement accuracy. This helped validate the precision of the UAV's environment-sensing capability, which is critical for safe navigation in constrained spaces.

For moving object detection and response time, the HB100 Doppler Radar was employed to detect dynamic obstacles such as a rotating fan or a second UAV in motion. The radar's response time—measured as the interval between the object's movement and the UAV's initiation of an avoidance manoeuvre—was recorded. This metric is vital in ensuring that the UAV can react swiftly and effectively to unpredictable, fast-approaching hazards in real-time.

In the flight path correction and manoeuvring tests, the UAV was placed in controlled environments requiring real-time directional changes based on incoming obstacle data. The system logged the time taken to complete each manoeuvre and recorded the success rate of obstacle avoidance. This provided insight into the UAV's agility and decision-making effectiveness when navigating complex airspace.

Finally, the real-time data transmission test focused on evaluating the Joystick module's performance in delivering obstacle alerts to a connected Android application. The system was assessed for latency in data transmission and overall signal reliability. Ensuring fast and reliable communication between the UAV and user interface is essential for monitoring, control, and enhancing situational awareness during flight operations. Together, these evaluations confirmed the

robustness and responsiveness of the UAV's integrated detection and avoidance system.

6.3.1 Test Scenarios

6.3.1.2 Static Obstacle Avoidance Test

Verifying the UAV's ability to detect and avoid stationary obstacles. The UAV is flown toward a wall or large object at different speeds. Expected Outcome: The UAV detects the obstacle and executes yaw (turn left/right) or altitude change.

6.3.1.3 Moving Obstacle Avoidance Test

Test the response of HB100 Do

ppler Radar to a moving obstacle. A moving object (e.g., drone, swinging pendulum) is introduced in the UAV's path. The UAV detects the motion, calculates velocity, and either stops, hovers, or redirects flight path.

6.3.1.4 Communication and Data Transmission Test

Test Joystick-based real-time monitoring of obstacle detection alerts. The UAV transmits sensor data to Joystick module . No significant delay in data transmission.

6.4 Performance Metrics for UAV Collision Avoidance System

6.4.1 Obstacle Detection Accuracy

The obstacle detection accuracy measures how accurately the UAV detects and measures obstacles compared to actual distances. The accuracy obtained was $\geq 95\%$. Waveforms were captured for the case of without obstacle and with obstacle using an oscilloscope.

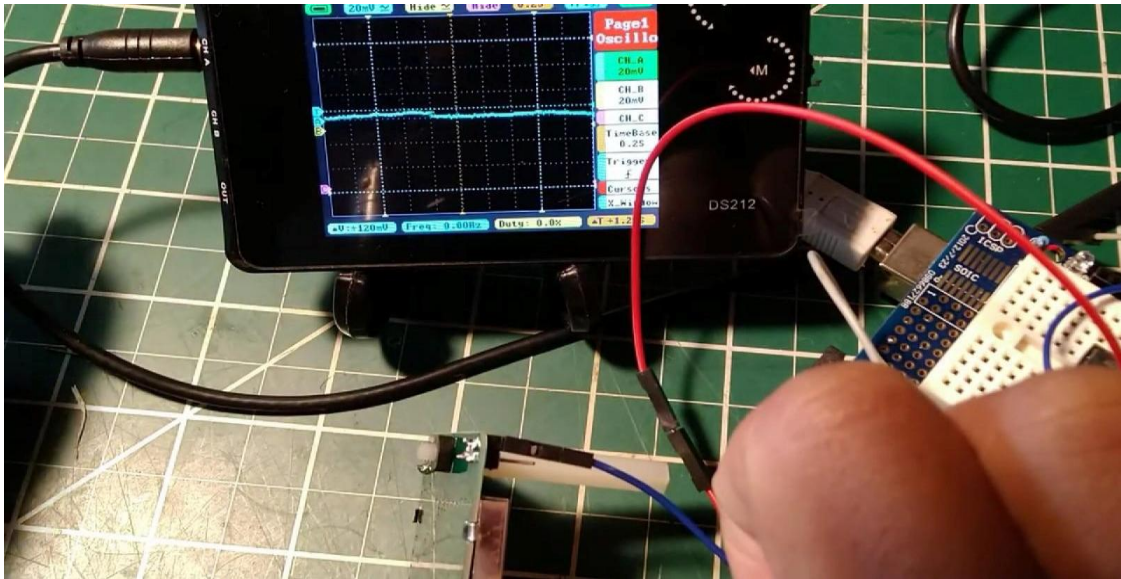


Figure 6.3 Output waveform of the hardware when there is no Obstacles

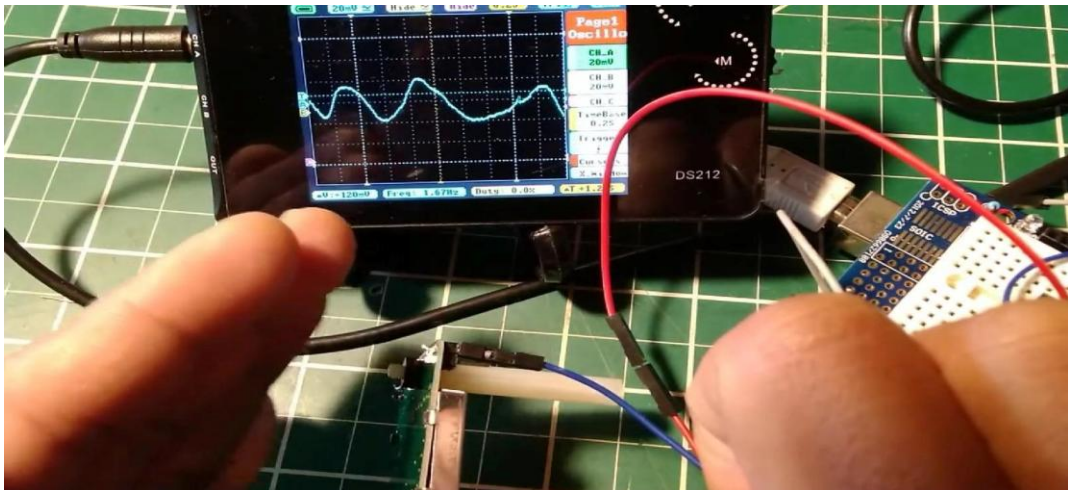


Figure 6.4 Output wave form of the hardware when there is Obstacles detected

Figure 6.3 illustrates the waveform pattern observed when no obstacle is detected by the UAV's sensors. In this state, the waveform remains steady and consistent, indicating a clear path with no interference or reflection caused by nearby objects. The LiDAR or radar signals return smoothly without sudden spikes or disturbances, reflecting stable surroundings and uninterrupted sensor readings.

In contrast, Figure 6.4 displays the waveform pattern when an obstacle is detected. Unlike the flat waveform in Figure 6.3, this graph shows noticeable spikes or irregularities in the signal, caused by the reflection of the sensor waves from nearby

obstacles. These variations indicate a change in the environment, triggering the UAV's obstacle avoidance algorithm. The shift in waveform confirms the presence of an object within the detection range, prompting immediate corrective actions such as slowing down, stopping, or manoeuvring around the obstacle.

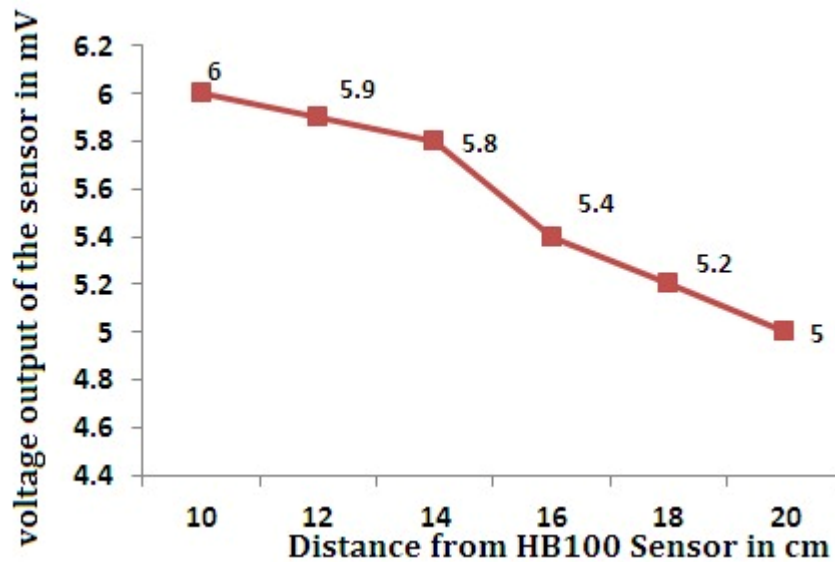


Figure 6.5 Distance vs Voltage output of HB100 sensor

Test was conducted to analyse the working of HB100 sensor. Different test conditions was used to test its performance for different distance. Figure 6.5 shows the variation of the output voltage for the variation in the distance from the sensor. It is found that as the distance from the obstacle is decreased, the voltage output is higher.

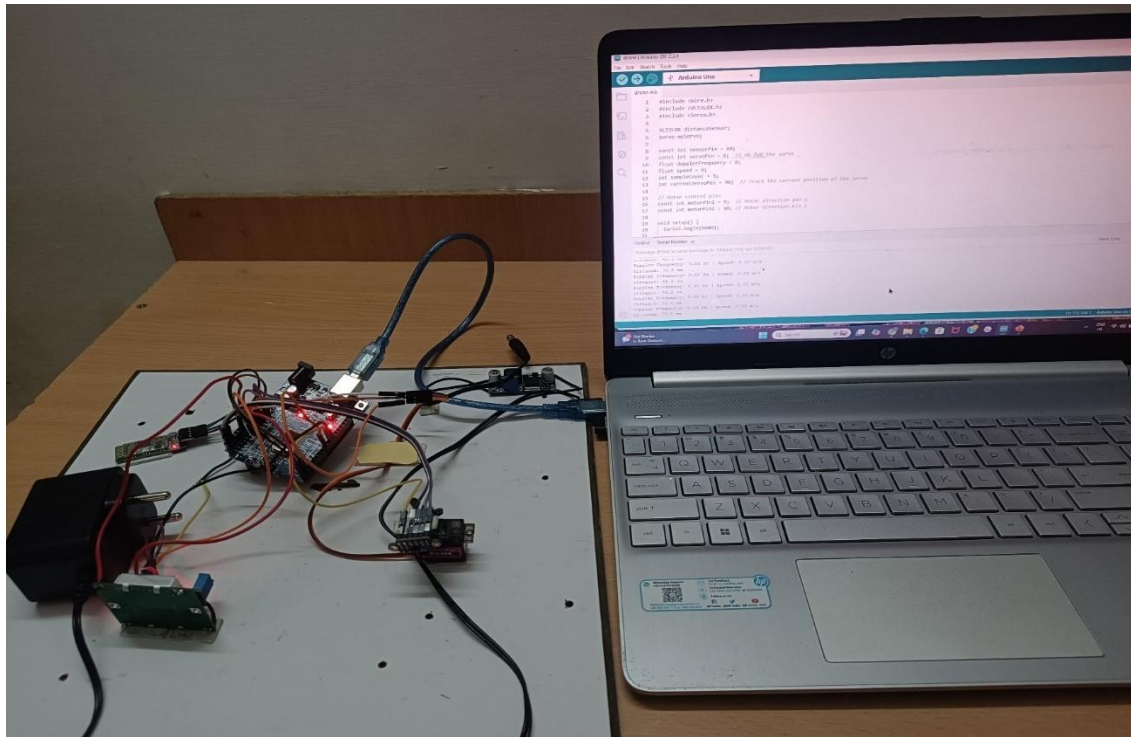


Figure 6.6 Screen shot of the proposed UAV system

Figure 6.6 shows integrated hardware module consists of hardware components and connected with simulation tool to get the results and figure 6.7 shows the display of the frequency, speed and distance in OLED in the hardware setup.

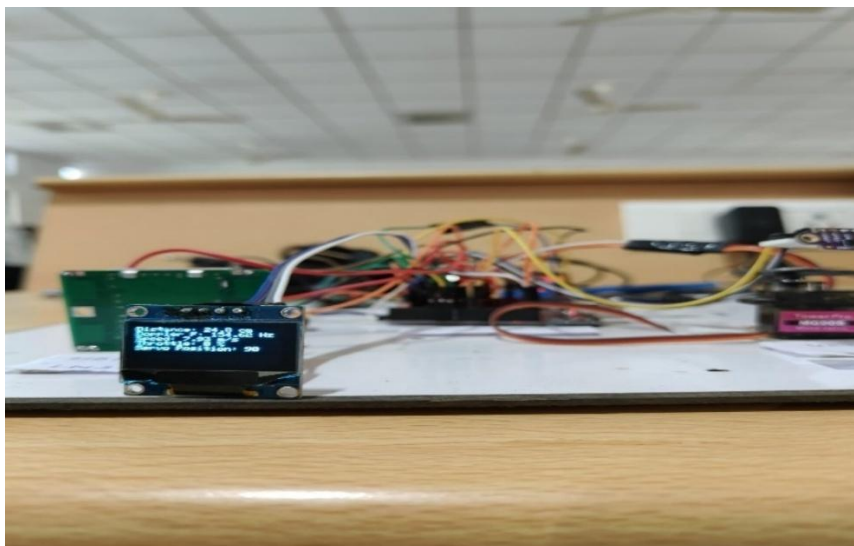


Figure 6.7 Display of frequency, speed and distance in OLED

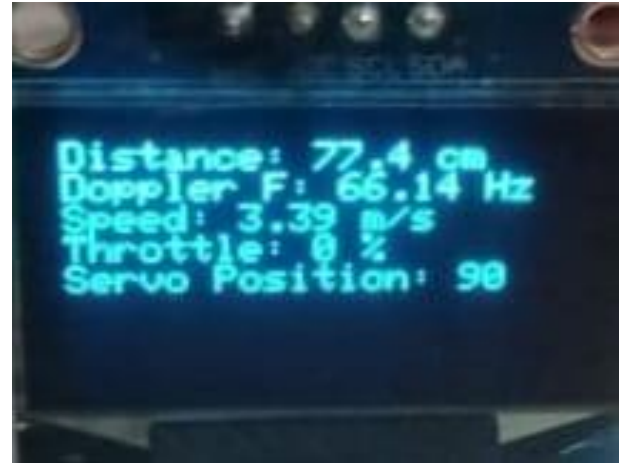


Figure 6.8 OLED display for different distance detected by sensors

Figure 6.8 shows that the output of sensors for different distance and frequency. IT is inferred that the speed is high for the distance between the UAV and the obstacle is high.



Figure 6.9 OLED display for different distance and throttle

Figure 6.9 shows the output of sensors for different throttle and distance. From the figure it is found that the throttle is high for the distance between the obstacle and drone to be larger. When the distance decreases and is equal to the threshold value, the throttle cannot be increased and the drone will not be moving further or hovering. Thus, throttle has its impact till a minimum distance.

Conclusion

The system integrates multiple sensors to enable autonomous and manual control of a UAV. The VL53L0X time-of-flight sensor detects static obstacles and transmits distance data via I2C, which is then processed and displayed on an OLED screen alongside decision logic for navigation. For dynamic obstacle detection, an HB100 Doppler radar sensor coupled with an LM358 amplifier outputs an analog signal to pin A0, triggering servo motors to adjust direction accordingly. A joystick module connected to pins A2 and A3 provides manual override and directional input, controlling motor speed and direction when necessary. Servo motors respond to either sensor-based avoidance logic or manual commands to steer the UAV. The OLED display continually updates with real-time information on distance measurements, motion detection status, and control mode, ensuring effective monitoring and decision-making.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

Conclusion:

The project that we are proposed that will detect the obstacle and collision avoidance using sensors. The HB100 sensor, VL53L0X sensor, Arduino Uno, Joystick module, and OLED display were all used in the successful implementation of the Obstacle Detection and Throttle Control System. With the VL53L0X sensor providing precise distance readings and the HB100 sensor detecting obstacles by altering its frequency, the project was able to accomplish real-time obstacle detection and throttle adjustment. Keys tricks were used to precisely control the throttle up to 100%, Real-time feedback from the OLED displays enhanced monitoring and user interaction. To maximize performance, a few minor issues with sensor calibration and Joystick denounce handling were successfully resolved. With its responsive, dependable, and adaptable obstacle detection and throttle control system, the system achieved its goals and has potential uses in robotics, autonomous navigation, and smart

Future Scope:

Autonomous navigation with accurate localization and motion tracking would be made possible by integrating GPS and IMU (Inertial Measurement Unit) sensors. The system can differentiate between various obstacles, including moving cars, pedestrians, and stationary objects, by introducing machine learning algorithms for dynamic obstacle classification. The system would be better suited for cutting-edge applications in industrial robotics, agricultural automation, and driverless cars if Wi-Fi and Internet of Things-based communication were added. This would allow for real-time monitoring and control from remote locations.

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